MID / LOW-LEVEL DESIGN

Windows Admin Center for Azure Local at Scale

Three first-class deployment patterns for Windows Admin Center (WAC) when managing many Azure Local instances across regions and sites. Pick the pattern that fits the operating model; the security, network, OT and lifecycle baseline is the same across all three.

Audience: Platform / Cloud / OT engineering

Sources: Microsoft Learn only

Scope: Production & Dev/Test

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1. Executive summary

Microsoft supports several ways to deploy WAC for managing Azure Local. This document presents the three patterns customers typically choose between, each at a regional scale (one HA pair or service surface per geography — never one WAC per cluster). All three include a per-site break-glass WAC for local-only operations during WAN or Azure outages.

Pick the pattern that aligns with your operating model and connectivity assumptions; the network, security, OT and lifecycle baseline in §4 applies to all three.

Pattern A

On-prem HA WAC (regional)

HA WAC on a regional management cluster, per geo. Best fit when sites must keep full management during WAN or Azure outages and OT requires east-west-only management traffic.

Go to Pattern A →

Pattern B

WAC in Azure VMs (regional HA)

HA WAC on Azure VMs per geo, reaching sites over ExpressRoute or S2S VPN. Best fit when an ARM-first operating model and minimal on-prem management footprint are the priority and private connectivity to every site already exists.

Go to Pattern B →

Pattern C

WAC in Azure portal (service)

The in-portal WAC experience — currently in **preview for Azure Local** cluster nodes. Useful today for opportunistic Arc-server tasks; a candidate for primary use once GA and at feature parity with the on-prem gateway.

Go to Pattern C →

ALL PATTERNS INCLUDE

A **per-site break-glass WAC** (Tier 2): a small Windows Server VM at each site, powered off by default, activated only when WAN/ER to the regional WAC is unavailable or when an OT-approved local-only operation is required. See §4.1.

2. Terminology & gateway-mode disambiguation

Three distinct things are commonly called "gateway" in this space; lock the definitions down before discussing topology:

Term	What it actually is	Where it lives
WAC gateway server	An installation mode of Windows Admin Center where the WAC `.exe` is installed on a Windows Server (or on a failover cluster for HA), and admins connect to it from a browser. Used by Patterns A and B.	Windows Server (single VM or 2-node cluster), on-prem (Pattern A) or in Azure (Pattern B).
Azure Arc gateway for Azure Local	An *Azure* resource that consolidates the outbound endpoints Azure Local nodes need to reach Azure. Reduces firewall allow-list size. Has nothing to do with WAC.	Azure (resource) + Arc proxy agent on each Azure Local node.
WAC in Azure portal	An Azure portal blade that lets you run a WAC-style UI against Arc-enabled servers, Azure Local nodes, and Azure VMs without an inbound network path. Documented as *preview* for Azure Local cluster nodes. This is Pattern C.	Azure portal (no on-prem server install).

Other terms used in this document

- **Tier 1** — the regional, day-to-day management surface (Pattern A's HA pair, Pattern B's Azure VMs, or Pattern C's portal blade).
- **Tier 2** — the per-site break-glass WAC, common to all three patterns.
- **ER** — Azure ExpressRoute. **S2S** — Site-to-Site VPN.
- **ILB** — Azure Standard Internal Load Balancer.
- **Run mode** — how Tier 1 is operated: Always-on HA, Cold-standby (deallocated, started on demand), or Auto-shutdown nightly (Dev/Test only). See §4.2.
- **JIT (just-in-time) start** — a Cold-standby variant where a runbook starts the deallocated VMs on operator request or alert.

> **PLAIN ENGLISH**
> WAC gateway = the box you install. Arc gateway = an Azure-side traffic concentrator. WAC in Azure portal = a managed UI with no install. They are independent and can coexist.

3. Pick your pattern

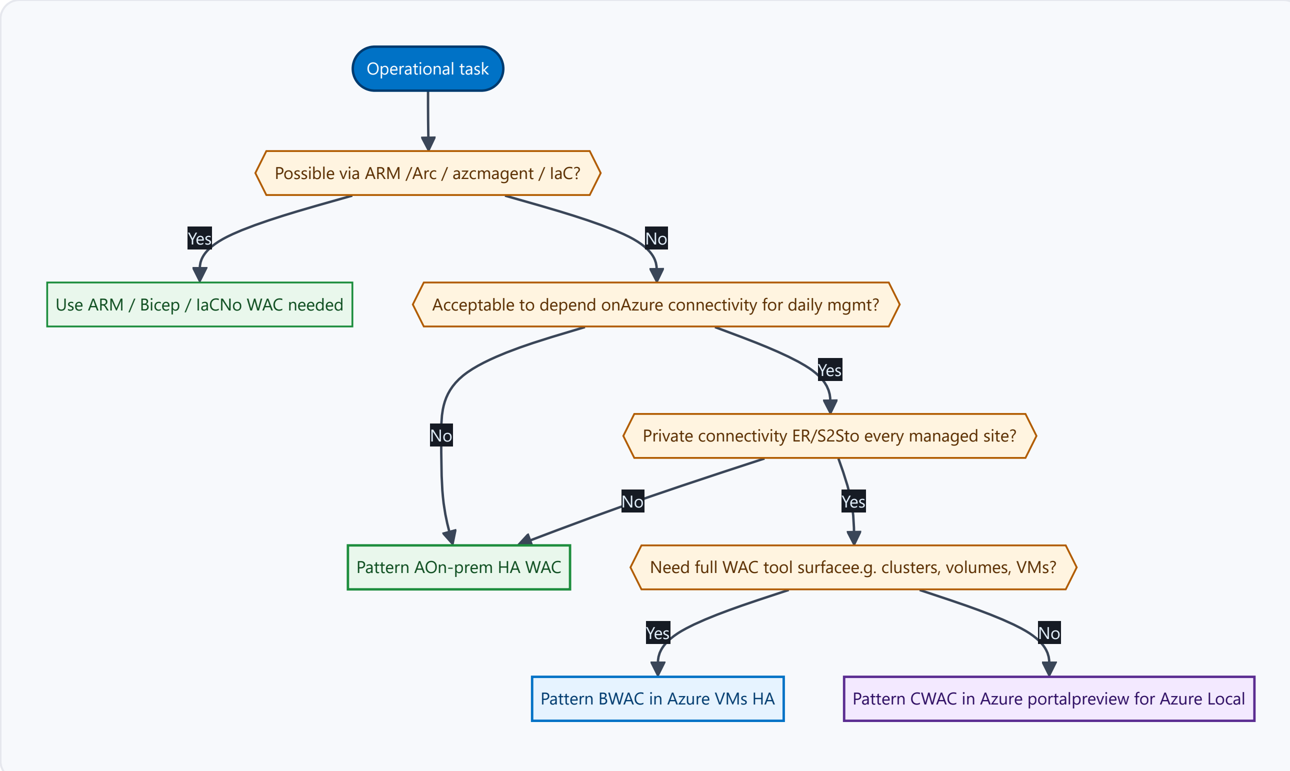
Each pattern below is a valid Microsoft-supported way to deploy WAC at regional scale. The right choice depends on connectivity assumptions, OT segmentation requirements and the desired split between on-prem and Azure footprint. None of the three is "wrong" — they are operating-model choices.

3.1 Comparison

Dimension	A On-prem HA	B Azure VMs HA	C Azure portal
Tier 1 location	Regional on-prem mgmt cluster	Azure VMs per geo (HA pair behind ILB)	Azure portal (no Tier 1 footprint)
Connectivity prerequisite	Site mgmt VLAN routing only	ExpressRoute or S2S VPN to every managed site	Sites Arc-enabled and able to reach Azure
Behaviour during WAN / Azure outage	Tier 1 unaffected; sites still managed	Tier 1 unreachable; Tier 2 takes over per site	Tier 1 unreachable; Tier 2 takes over per site
OT / segmentation profile	Mgmt traffic stays east-west inside the site	Mgmt traffic enters site over private circuit from Azure	Mgmt via Arc agents; no inbound flow to site
On-prem footprint	Highest (regional cluster + site break-glass)	Low (site break-glass only)	Lowest (site break-glass only)
Azure footprint	None for Tier 1	2× VMs + ILB + Key Vault per geo	Service only — no resources to manage
Cost profile (steady state)	CapEx-heavy (cluster), OpEx low	OpEx scales with run mode (see §4.2)	Service cost only; ARM operations counted
Run modes available	Always-on HA	Always-on / Cold-standby / JIT	N/A (managed service)
Supportability today	GA	GA	Preview for Azure Local cluster nodes
Best fit when…	Sites must retain full mgmt during WAN/Azure loss; OT mandates east-west mgmt only	Private connectivity to every site exists; ARM-first ops; minimal on-prem mgmt VMs preferred	Zero on-prem mgmt VMs preferred; Arc-server day-to-day is enough until in-portal WAC reaches GA / parity

All three options are documented Microsoft installation patterns (see references in §10). The Tier 2 break-glass is identical across patterns — see §4.1.

3.2 Decision tree



All paths assume a per-site break-glass WAC (Tier 2) is also in place — see §4.1.

4. Shared baseline (applies to all patterns)

The content in this section is identical for Patterns A, B and C. Each pattern chapter (§5–§7) references back here.

4.1 Per-site break-glass WAC (Tier 2)

One small Windows Server VM per site, on the site's management VLAN — a regular workload VM, *not* a cluster node, *not* in Azure — and powered off by default. Activated only when the regional Tier 1 (Pattern A, B or C) is unreachable or when an OT-approved local-only operation is required.

4.1.1 Where it lives

1. **As an Azure Local VM on the site's own cluster** (recommended). It is a guest workload, not a cluster node. Survives WAN / Azure loss because Azure Local continues to operate while disconnected from Azure (within the documented Arc grace period).
2. **On a dedicated small management host** at the site, if one already exists. Useful where local standards require the management plane to be physically separate from the workload cluster.

DO NOT

> Do not host Tier 2 in Azure, on the Azure Local cluster *nodes* themselves, on a domain controller, or on the OT process network.

4.1.2 Build

- Windows Server 2022/2025, 2 vCPU / 4 GB / 127 GB. Domain-joined to the local site AD (or workgroup if AD is unavailable at the site).
- WAC installed in *gateway server* mode on port 443. Enterprise CA cert preferred; otherwise a self-signed cert with documented thumbprint trust on the operator workstation.
- Local admin break-glass account documented in the site's privileged-access vault. Avoid a hard dependency on Entra ID for the activation path.
- Keep the WAC version within one release of the regional Tier 1 to minimise behavioural drift between tiers.

4.1.3 Activation runbook

WHY NOT JUST LEAVE IT ON?

> Leaving Tier 2 always-on doubles the attack surface per site and introduces drift between Tier 1 and Tier 2 RBAC. Powered-off-by-default keeps the management segment quiet under normal operation and forces a deliberate, logged activation.

4.2 Run modes

"Run mode" is how Tier 1 is operated day-to-day. Pattern A only supports Always-on HA (it is a clustered Windows install). Pattern B can use any of the three run modes; the choice is a cost-vs-RTO trade-off. Pattern C is a managed service and has no run mode in this sense.

Run mode	How it works	RTO	Cost shape	Applies to
Always-on HA	Both Tier 1 nodes are running 24×7. Active-passive failover under load balancer or cluster client access point.	Seconds (failover) — operator is unaffected by single-node loss.	Highest — full compute billed continuously.	Patterns A and B.
Cold-standby / JIT	Tier 1 VMs are deallocated when not in use. Started by a runbook (Logic App, Automation Account or ARM template) on operator request, alert, or scheduled window.	~5–10 minutes (boot + cert + extension warm-up).	Low — only storage billed at rest; compute billed only when started.	Pattern B only (Azure VMs can be deallocated; on-prem cluster nodes cannot meaningfully be).
Auto-shutdown nightly	VM auto-shuts down outside business hours, auto-starts on a schedule.	~5 minutes during shutdown window.	Low — only billed during business hours.	Dev/Test only — not appropriate for production Tier 1.

> **JIT PATTERN OUTLINE**
> A typical JIT setup: an Azure Automation runbook (or Logic App) takes a parameter — a region — and starts the two Tier 1 VMs in that region; an operator triggers it via a self-service URL fronted by Entra ID + Conditional Access. After a configurable idle period the runbook deallocates them again. Microsoft Learn references for VM start/stop automation are listed in §10.

4.3 Network, firewall & OT requirements

4.3.1 WAC gateway → Azure Local cluster nodes (east-west, on-prem)

Source: Microsoft Learn — Firewall requirements for Azure Local.

Purpose	Source	Destination	Protocol	Port
SMB (file ops, cluster validate)	WAC	Azure Local nodes	TCP	445
WinRM HTTP (PowerShell remoting)	WAC	Azure Local nodes	TCP	5985
WinRM HTTPS (preferred)	WAC	Azure Local nodes	TCP	5986
RPC Endpoint Mapper	WAC	Azure Local nodes	TCP	135
Failover Cluster service	WAC	Azure Local nodes	TCP/UDP	3343
RPC dynamic (Failover Cluster, Hyper-V VMM)	WAC	Azure Local nodes	TCP	≥ 5000 (open ≥ 100 ports)
Hyper-V Live Migration	Mgmt system	Hyper-V hosts	TCP	6600
Hyper-V VM Management	Mgmt system	Hyper-V hosts	TCP	2179

> **TIP**
> If WAC is installed with the "Use WinRM over HTTPS only" option, allow only 5986 and skip 5985. In Pattern B these flows traverse the ER/S2S circuit from Azure into the site management VLAN; in Pattern A they stay east-west inside the site.

4.3.2 WAC gateway → Internet (north-south)

Source: Microsoft Learn — WAC network requirements. All HTTPS / TCP 443.

FQDN	Purpose	When required
`aka.ms`	Acquiring & maintaining WAC	Always
`download.microsoft.com`	Acquiring & maintaining WAC	Always
`pkgs.dev.azure.com`	Extension management	Always
`*.vsblob.visualstudio.com`	Extension management	Always
`login.microsoftonline.com`	Entra ID auth	When using Azure Hybrid Services (Public)
`graph.microsoft.com`	Microsoft Graph	When using Azure Hybrid Services (Public)
`graph.windows.net`	AAD Graph (legacy)	When using Azure Hybrid Services (Public)
`management.azure.com`	ARM	When using Azure Hybrid Services (Public)

4.3.3 Browser → WAC and browser → Azure

FQDN	Purpose	When required
`winadmincenterassets.blob.core.windows.net`	Extension management	Always
`js.monitor.azure.com`	Extension management	Always
`nuget.org`	Extension management	Always
`announcements.blob.core.windows.net`	Extension management	Always
`browser.events.data.microsoft.com`	Telemetry	Optional
`login.microsoftonline.com`, `graph.microsoft.com`, `graph.windows.net`, `portal.azure.com`	Azure Hybrid Services	When used

4.3.4 Arc gateway for Azure Local — orthogonal to WAC

The **Azure Arc gateway** for Azure Local is unrelated to WAC topology but materially reduces the Azure-outbound endpoint list each Azure Local node needs. Key facts (Microsoft Learn):

- Enable Arc gateway **during deployment** of new Azure Local instances on version 2506+. It cannot be enabled post-deployment today.
- Arc gateway does **not** support TLS-terminating proxies.
- A short list of endpoints (e.g. `aka.ms`, `login.microsoftonline.com`, `management.azure.com`, `<region>.his.arc.azure.com`, `<arcgwid>.gw.arc.azure.com`, Key Vault, Cloud Witness, CRL/OCSP) is *not* redirected and must remain in the firewall allow-list.
- Azure Arc Private Link Scopes are not supported by Azure Local; Arc endpoints (`*.his.arc.azure.com`, `*.guestconfiguration.azure.com`, `*.dp.kubernetesconfiguration.azure.com`) must resolve to public IPs from the nodes / ARB / proxy.

> **NETWORK — MUST READ**
>
> **HTTPS / TLS inspection is not supported** for Azure Local outbound traffic. This includes Entra ID tenant restrictions v1. Coordinate with the network team early — this is the single most common deployment blocker on regulated estates.

4.3.5 OT segmentation pattern (recommended)

- WAC gateway servers (Tier 1 in any pattern, plus Tier 2) sit in the same management VLAN as the Azure Local cluster's management network — never on the OT process network and never on the storage / RDMA fabric.
- Operator browser traffic enters via a documented jump path (PAW, VDI, or Bastion in Pattern B) → Tier 1 client access point only.
- Tier 2 has no inbound from corp WAN; only the local site PAW reaches it.
- Pattern A: WAC→node traffic stays inside the site mgmt VLAN; only Azure-outbound (443) traverses the site firewall, ideally via the Arc gateway.
- Pattern B: WAC→node traffic enters the site mgmt VLAN over the ER/S2S circuit — coordinate with the network team to scope the source to the Tier 1 VNet only.

4.4 Security baseline

Control	Setting
Identity provider (Tier 1)	Microsoft Entra ID + Conditional Access + MFA (smartcard optional via AD).
Identity provider (Tier 2)	Local AD / local groups; documented break-glass accounts in privileged vault.
Gateway access roles	Separate *gateway administrators* from *gateway users*.
Target-node RBAC	WAC RBAC (JEA endpoint) — Administrators / Readers / Hyper-V Administrators built-in roles.
Certificates	Enterprise CA, private key on every HA node, rotated via `-UpdateCertificate`. In Pattern B, store certs in Azure Key Vault and reference from the VMs.
Where WAC may NOT be installed	Domain controllers — unsupported per Microsoft Learn.
Local management of self	Don't use WAC to manage the same server it's installed on.
Auditing	WAC logging enabled; forward to SIEM (e.g. Microsoft Sentinel) via Windows Event Forwarding or AMA where supported.

4.5 Operations & lifecycle

- **Patching:** WAC non-preview versions are supported until 30 days after the next non-preview release. Stay within one release of current.
- **Version skew:** 2311 → 2410+ requires uninstall + reinstall on HA. Plan a maintenance window. Applies to Patterns A and B.
- **Cluster Manager extension:** ensure version ≥ **5.2.6** on every WAC instance before performing volume operations on Azure Local; otherwise update to WAC 2511 build 2.6.6.18 or higher (or use WAC-in-Azure-portal extension 0.76.0.0+). Source: WAC known-issues, April 2026 update.
- **Tier 2 drift control:** quarterly job to power on each break-glass VM, patch, validate, power off. Track in change management.
- **Backup:** Pattern A — back up the CSV and the cert. Pattern B — protect via Azure Backup (VM-level) and Key Vault soft-delete on the cert. Tier 2 is stateless enough that re-imaging is the recovery path.

4.6 VM console / out-of-band access

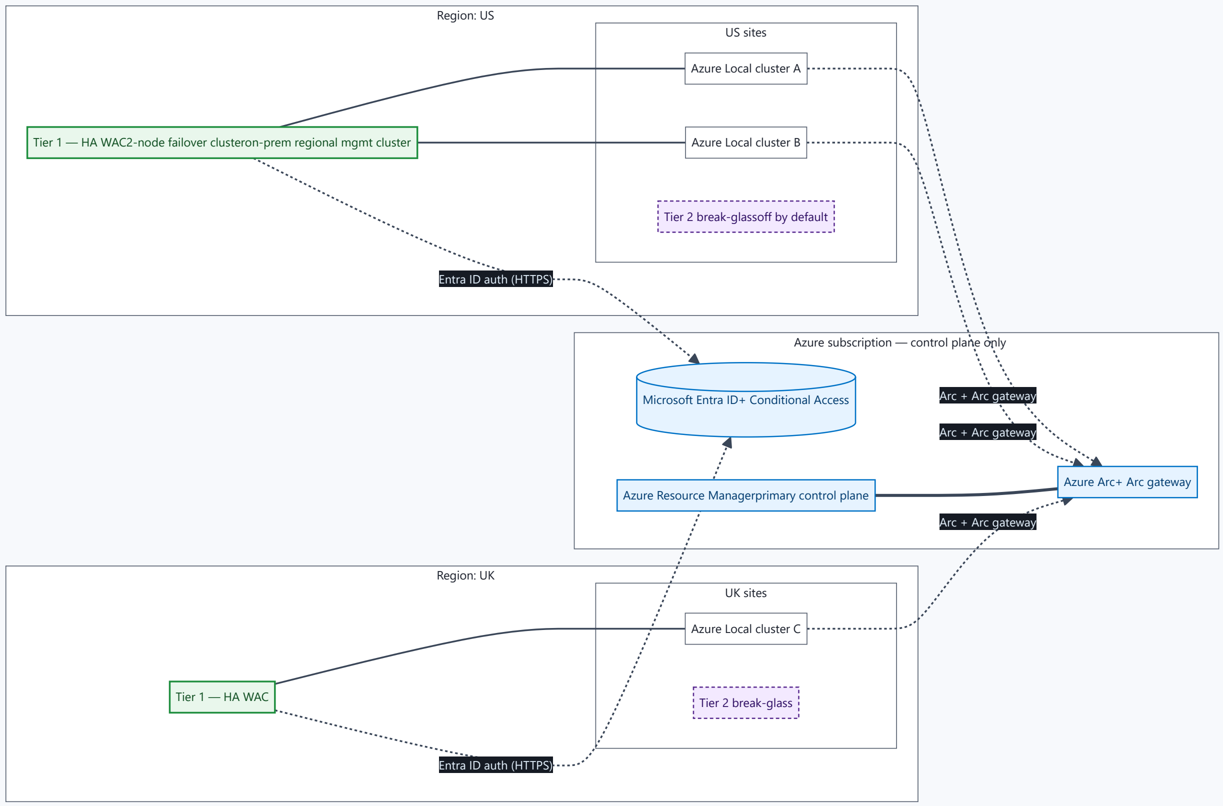
WAC does **not** solve the "I need a console when WAC itself is broken" problem. Treat this as a separate, parallel capability and document it in the same runbook:

- OEM BMC (iDRAC / iLO / XClarity) on a dedicated, isolated mgmt VLAN, reachable only from a site PAW.
- Hardened jump host per site for serial-over-LAN / KVM access.
- Document MFA and credential rotation for BMC accounts; these are typically the most-overlooked privileged accounts in regulated estates.

Pattern A 5. On-prem HA WAC (regional)

One HA WAC pair per geography (e.g. US, UK, APAC) running on a regional Windows Server failover cluster, with a per-site break-glass WAC at every site (§4.1).

5.1 Architecture



5.2 Prerequisites

- 2-node Windows Server failover cluster (Windows Server 2022 or 2025 recommended).
- One Cluster Shared Volume (CSV), ~10 GB, for WAC persistent data.
- TLS certificate from a trusted CA, with the private key installed on *every* node; capture the thumbprint.
- WAC HA scripts: aka.ms/WACHAScript (`Deploy-GatewayV2Ha.zip`).
- WAC installer (`WindowsAdminCenter<version>.exe`) staged on a node.
- DNS A-record for the chosen client access point (e.g. `wac-eu.contoso.local`) and an optional static IP for the cluster generic service.

5.3 Sizing (starting point)

The HA install is **active-passive** per Microsoft Learn — one node serves traffic at a time and the other takes over on failure. Size each node for full peak load.

Resource	Per node	Notes
vCPU	4 (suggested starting point)	Modernised gateway is multi-process; scale up for many concurrent admins.
Memory	8–16 GB (suggested)	Increase if many extensions are loaded.
OS disk	127 GB (suggested)	Standard Windows Server image.
CSV (shared)	≥ 10 GB	Per Microsoft Learn HA prerequisites.
Network	1× mgmt NIC (+ optional cluster NIC)	Same VLAN as the Azure Local management network or routed to it.

Microsoft Learn does not publish prescriptive vCPU/RAM minimums for the modernised gateway; values above are pragmatic starting points for review.

5.4 Install the HA gateway

From a node in the cluster, with the script and installer locally:

```
$parameters = @{
  ClusterStorage          = "C:\ClusterStorage\Volume1\Gateway"
  ClientAccessPoint       = "wac-eu"          # becomes https://wac-eu.<domain>
  StaticAddress           = "10.10.0.50"      # optional
  InstallerPath           = "C:\Installers\WindowsAdminCenter2511.exe"
  CertificateThumbprint = "AA11BB22CC33DD44EE55FF66AA77BB88CC99DD00"
}
.\Deploy-GatewayV2Ha.Deploy.ps1 @parameters
```

Validate and inspect with the sibling scripts (`Deploy-GatewayV2Ha.Validate.ps1` , `Deploy-GatewayV2Ha.Inspect.ps1`). To upgrade WAC later, re-run the same `Deploy-GatewayV2Ha.Deploy.ps1` with the new `-InstallerPath` .

> **UPGRADE CAVEAT**
> Direct upgrades from WAC HA **2311 and older to 2410 and newer are not supported** due to architectural changes in the modernised gateway. Plan an uninstall + reinstall when crossing that boundary.

5.5 Run modes

Pattern A supports **Always-on HA** only — the failover cluster expects both nodes online for HA to function. See §4.2.

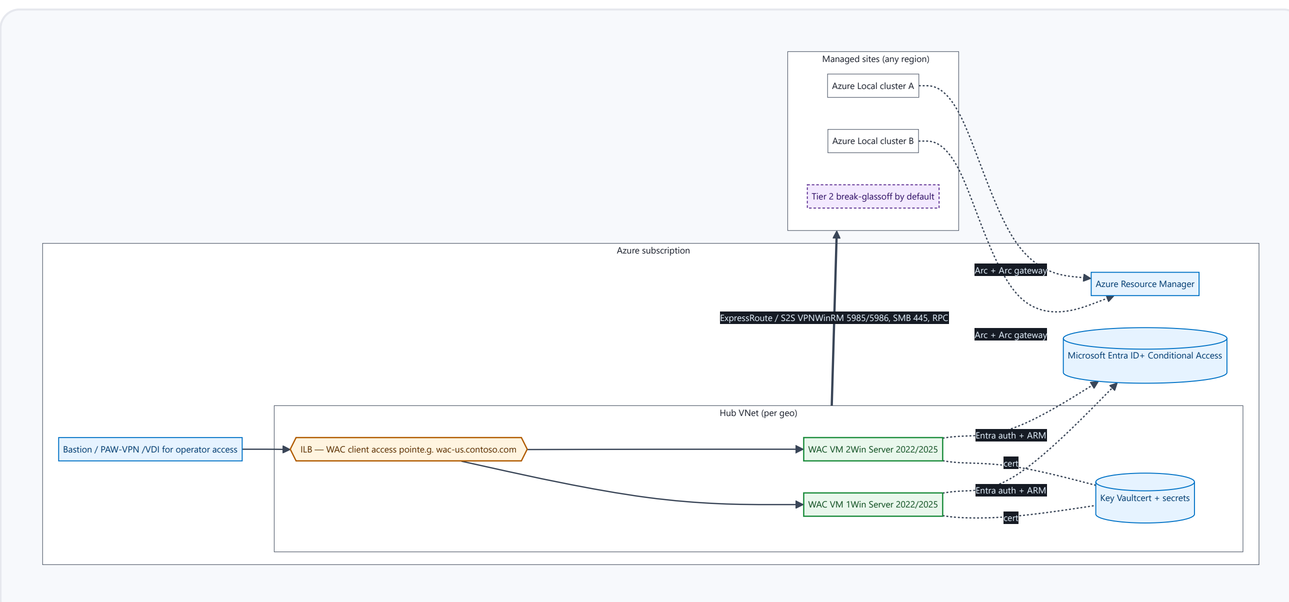
5.6 Trade-offs

Pros	Cons
• Tier 1 is unaffected by WAN or Azure outages. • Management traffic stays east-west inside the site — easiest profile for OT and network teams to approve. • No Azure egress dependency for daily management.	• Requires a regional management cluster (CapEx + ongoing patch / cert lifecycle). • Adds an on-prem failure domain to operate. • Run-mode flexibility limited to Always-on (no cold-standby option).

Pattern B 6. WAC in Azure VMs (regional HA)

One HA WAC pair per geography running on Azure VMs, fronted by an Azure Standard Internal Load Balancer (ILB), reaching managed sites over ExpressRoute or Site-to-Site VPN. Per-site break-glass WAC at every site (§4.1).

6.1 Architecture



6.2 Prerequisites

- **Private connectivity to every managed site** — Azure ExpressRoute (private peering) or Site-to-Site VPN. Pattern B is not appropriate for sites reachable only over the public internet.
- One hub VNet per geography hosting the Tier 1 WAC pair, with a dedicated subnet for the WAC VMs and a separate subnet for the ILB.
- Azure Key Vault in the same region for the WAC TLS certificate and any service-principal secrets.
- NSGs locking inbound to the WAC subnet to the operator-access subnet (Bastion / PAW-VPN / VDI) only.
- DNS A-record for the WAC client access point (resolves to the ILB front-end IP).
- Site firewall scoped to accept the WAC management flows (§4.3.1) sourced from the WAC subnet only.

6.3 Sizing (starting point)

Resource	Per VM	Notes
SKU	D4s_v5 (4 vCPU / 16 GB)	Starting point; right-size based on operator concurrency.
OS disk	Premium SSD 127 GB	Standard Windows Server image.
NIC	1× private NIC in WAC subnet	No public IP. Accelerated networking on.
Availability	Availability Zones (one VM per zone)	Required for the platform SLA on the ILB front-end.
ILB	Standard, internal, zone-redundant	Health probe on TCP 443 to each VM.

6.4 Build outline

1. Provision the hub VNet, subnets, NSGs and ILB (Bicep / Terraform / portal).
2. Deploy two Windows Server 2022/2025 VMs into different Availability Zones in the WAC subnet.
3. Domain-join (or stay workgroup with Entra-only auth — both are supported).
4. Install WAC in *gateway server* mode with the certificate retrieved from Key Vault.
5. Add the ILB back-end pool entries and the TCP/443 health probe.
6. Configure Entra ID auth + Conditional Access + MFA for the WAC tenant app.
7. Open WAC → site flows (§4.3.1) from the WAC subnet only.
8. Test failover by stopping each VM in turn; the ILB should bring traffic to the survivor within seconds.

> **NOTE**
> Pattern B uses **two independent VMs behind ILB** rather than the WAC HA failover-cluster script. The HA failover-cluster pattern (Pattern A) is designed for on-prem failover clustering; in Azure, ILB-fronted active-passive across Availability Zones is the cleaner shape and supports all three run modes.

6.5 Run modes

Pattern B supports all three run modes from §4.2:

- **Always-on HA** — both VMs running 24×7. Recommended for estates with continuous operator demand.
- **Cold-standby / JIT** — VMs deallocated; started by an Automation runbook on operator request, alert or scheduled window. ~5–10 min RTO; pays only for storage at rest.
- **Auto-shutdown nightly** — only for non-production Tier 1 (e.g. a dedicated Dev/Test geography).

> **JIT FOR COST-AWARE ESTATES**
> If WAC is used a few hours per day for change windows, Cold-standby / JIT often cuts Tier 1 compute cost by 80%+ while still giving operators a managed self-service path to start the gateway. The runbook can also auto-deallocate after a configurable idle period.

6.6 Trade-offs

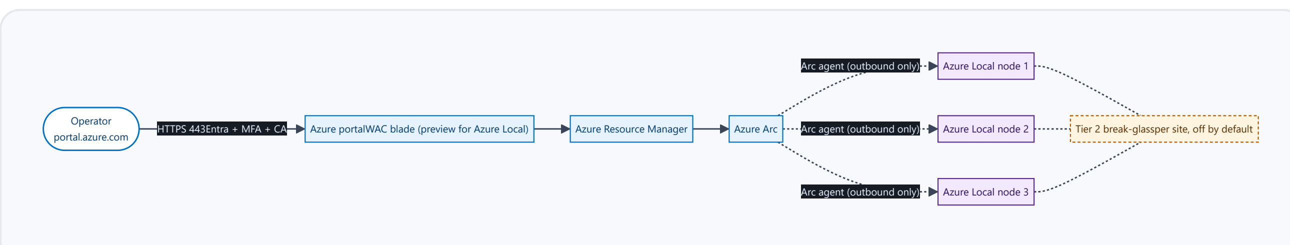
Pros	Cons
• Minimal on-prem management footprint (sites only need Tier 2 break-glass). • Native ARM-first ops: Bicep / Terraform deploy, Azure Backup, Key Vault, Azure Monitor. • Run-mode flexibility (Always-on / Cold-standby / JIT) for cost control. • Operator access path is Entra-fronted (Conditional Access, MFA, PIM).	• Requires private connectivity (ER or S2S VPN) to every managed site. • Management traffic crosses the WAN into the site mgmt VLAN — coordinate with OT / network teams to scope flows. • Tier 1 is unreachable during ER/S2S outages or Azure regional events; the per-site break-glass (Tier 2) covers this. • Latency on chatty WinRM operations vs. on-prem (typically tens of ms over ER) — usually acceptable, occasionally noticeable on large cluster operations.

Pattern C

7. WAC in Azure portal (service)

The in-portal WAC experience: an Azure portal blade running the WAC UI against Arc-enabled servers and Azure Local cluster nodes, with no WAC server to install, patch or run. Currently **preview** for Azure Local cluster nodes per Microsoft Learn. Per-site break-glass WAC at every site (§4.1) remains.

7.1 Architecture



7.2 Prerequisites

- Azure Local cluster nodes Arc-enabled (this is true by default for Azure Local 23H2+).
- Operator has the necessary Azure RBAC role on the Azure Local resource (typically Reader + a custom WAC role, or built-in roles such as Azure Connected Machine Resource Administrator depending on task).

- Outbound from each node to the documented WAC and Arc endpoints (see §4.3 — Arc gateway is the recommended way to consolidate this list).

7.3 What works today vs. what to track

Microsoft Learn documents the Azure Local cluster-node experience as *preview*. Treat the feature surface as a watch-list and revisit at each WAC release:

Capability	Status (as of doc date)
Arc-server day-to-day (services, files, registry, PowerShell, events)	GA via the Arc-server WAC blade.
Azure Local cluster node management	Preview per Microsoft Learn — feature parity with the on-prem gateway is still in flight.
Cluster Manager extension parity	Track the WAC release notes — extension version requirements differ between in-portal and on-prem (see §4.5).
VM management (Hyper-V tools)	Subset today; full toolset still in the on-prem gateway.

7.4 Run modes

N/A — this is a managed service. There is no Tier 1 VM to start, stop, deallocate or auto-shutdown. **Tier 2 break-glass at every site is still required** for outages where Azure / portal access is unavailable.

7.5 Trade-offs

Pros	Cons
• Zero on-prem management footprint, zero Tier 1 patching, zero certificate lifecycle. • Native Entra + Conditional Access + RBAC + audit trail in the Azure activity log. • Operator UX matches the rest of the Azure portal.	• Currently preview for Azure Local cluster nodes — not all WAC tools are at parity. • Full Azure dependency for the management plane during Azure / portal events; Tier 2 is the only fallback. • Site outbound to Azure must be intact; TLS inspection of those flows is unsupported (§4.3).

WHEN TO REVISIT

Pattern C is a strong candidate to *become* the primary Tier 1 once it reaches GA for Azure Local cluster nodes and parity with the on-prem gateway. Until then, use it for Arc-server tasks and keep Pattern A or B as the primary surface.

8. Dev/Test pattern

One Windows Server VM (*not* a domain controller — installing WAC on a DC is unsupported). Local AD or Entra ID for auth. Self-signed cert acceptable for short-lived labs.

Build

1. Provision Windows Server 2022/2025 VM (4 vCPU / 8 GB).
2. Domain-join to the lab AD (or workgroup).
3. Download the WAC installer from the Microsoft Evaluation Center.
4. Install in *gateway server* mode; bind to port 443.
5. Add lab admins to the *gateway administrators* group.

Parity vs. Production

- Same WAC tools and extensions are available.
- No HA / no failover — single point of failure (acceptable for lab).
- Use a *different* Entra ID app registration than production.
- Do not connect lab WAC to production Azure Local clusters.
- Auto-shutdown nightly is appropriate here (see §4.2).

9. Open questions for the customer

ANSWERS SHAPE THE CHOSEN PATTERN AND THE TIER 2 BUILD

1. Which pattern (A / B / C) is the target for Tier 1, and is that consistent across all geographies?
2. For Pattern B candidates: is private connectivity (ER or S2S VPN) in place to every managed site today? If not, when?

3. What OT / network control is being invoked (e.g. ISA/IEC 62443 zone separation, site-specific change-control rule, regulatory interpretation)? Tier 2 build differs for each.
4. Does each site have its own AD, or is AD homed regionally? Decides how Tier 2 authenticates with no WAN.
5. Confirm enterprise PKI source for WAC certificates and rotation cadence (and Key Vault use for Pattern B).
6. Confirm site count per region (drives the Tier 2 fleet size and patch rota).
7. Confirm whether Arc gateway will be enabled at *initial* Azure Local deployment (it cannot be added later).
8. Confirm operator workstation strategy (PAW vs. VDI vs. Bastion vs. corp laptop) — affects browser-side firewall rules and the access path.
9. For Pattern B Cold-standby / JIT: who owns the start/stop runbook and the self-service URL?

10. References (Microsoft Learn)

- Windows Admin Center overview
- What is Windows Admin Center?
- WAC — what type of installation is right for you?
- WAC — user access options
- Deploy WAC with high availability
- WAC — network requirements
- WAC — configure user access control
- WAC in the Azure portal — manage Azure VMs
- WAC in the Azure portal — manage Arc-enabled servers
- WAC — known issues
- Azure Local — firewall requirements
- About Azure Arc gateway for Azure Local
- Register Azure Local machines with Azure Arc gateway
- Azure Load Balancer overview (used in Pattern B)
- Azure Bastion overview (operator access path for Pattern B)
- Azure Key Vault overview (cert store for Pattern B)
- Start/Stop VMs during off-hours (run-mode automation)
- Azure ExpressRoute overview (Pattern B prereq)
- Azure VPN Gateway overview (Pattern B prereq)
- WAC HA deployment script (Deploy-GatewayV2Ha.zip)
- Download Windows Admin Center (Evaluation Center)

All links resolved against Microsoft Learn / Microsoft.com at time of writing. No third-party or non-Microsoft sources are used.

How to read this document. Self-contained HTML — open in Edge or Chrome. `Ctrl+P` → Save as PDF produces a clean, paginated copy (the left navigation is hidden in print). Diagrams render client-side via the Mermaid CDN (`cdn.jsdelivr.net`); if your workstation blocks third-party CDNs, request the offline-bundled version, which inlines the Mermaid library so the file works fully air-gapped.